



2.3) Fermi's golden rule

Non-relativistic case

Transition rate $T_{fi} = 2\pi |T_{fi}|^2 \rho(E_f)$

Transition matrix $\uparrow$

density of system $\rightarrow$

Hamiltonian $H = H_{\text{free}} + H_{\text{int}}$

$$T_{fi} = \langle f | H_{\text{int}} | i \rangle \quad \text{1st order transition}$$

○ T_{fi} is not Lorentz-invariant

We use M_{fi} the matrix element that is L.I.

○ Example of a particle decay $a \rightarrow 1 + 2$

Transition rate:

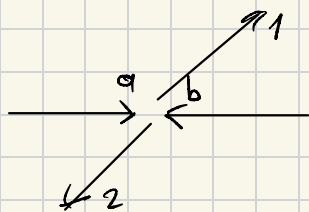
$$T_{fi} = \frac{(2\pi)^4}{2E_a} \int |M_{fi}|^2 \delta(E_a - E_1 - E_2) \delta^3(\vec{p}_a - \vec{p}_1 - \vec{p}_2) \times \frac{d^3\vec{p}_1}{(2\pi)^3 (2E_1)} \frac{d^3\vec{p}_2}{(2\pi)^3 (2E_2)} \Bigg\} \text{L.I. phase space}$$

○ Interaction cross-section

Interaction rate = Flux of incident particles $\times$ Number of particles in target $\times$ Cross-section σ

σ has the dimension of an Area!
unit: barn / b
 $1 \text{ b} = 100 \text{ fm}^2$

An example as illustration



$$\sigma = \frac{(2\pi)^2}{F} \int |M_{fi}|^2 \delta(E_a + E_b - E_1 - E_2) \delta^3(\vec{p}_a + \vec{p}_b - \vec{p}_1 - \vec{p}_2) \times \frac{d^3\vec{p}_1}{2E_1} \frac{d^3\vec{p}_2}{2E_2}$$

(flux)

Lorentz-invariant Flux $F = 4 \sqrt{(\vec{p}_a \cdot \vec{p}_b)^2 - m_a^2 m_b^2}$

3) Dirac equation

3.1) Reminders

Schrödinger eq. $E = \frac{\vec{p}^2}{2m} \xrightarrow{\text{QM operators}} \begin{cases} \hat{p} = \frac{\hbar}{i} \vec{\nabla} \\ \hat{E} = i\hbar \frac{\partial}{\partial t} \end{cases} \Rightarrow i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \vec{\nabla}^2 \psi$

Klein-Gordon eq. $E^2 = p^2 c^2 + m^2 c^4$

$$\Rightarrow \left(\vec{\nabla}^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right) \psi = \frac{m^2 c^2}{\hbar} \psi$$

- $\rightarrow$ It suggests negative probability densities for fermions
- $\rightarrow$ does not describe fermions
- $\rightarrow$ Works for scalar

3.2) Dirac equation

$$\hat{E} \psi = \hat{H} \psi = (\vec{\alpha} \cdot \vec{p} + \beta m) \psi$$

the solutions of this eq. must satisfy

$$E^2 = p^2 c^2 + m^2 c^4$$

$\Rightarrow$ the coefficient $\vec{\alpha}$ and β must then satisfy

$$\left\{ \begin{array}{l} \alpha_x^2 = \alpha_y^2 = \alpha_z^2 = \beta^2 = 1 \\ \alpha_j \beta + \beta \alpha_j = 0 \\ \alpha_j \alpha_k + \alpha_k \alpha_j = 0 \quad j \neq k \end{array} \right. \quad \text{Dirac Algebra}$$

One conventional choice of these coefficient is
Dirac-Pauli representation

$$\beta = \begin{pmatrix} \mathbb{1}_2 & 0 \\ 0 & -\mathbb{1}_2 \end{pmatrix}$$

$$\mathbb{1}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\alpha_i = \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix}$$

where σ_i are Pauli matrices

As a consequence ψ has 4 components

$$\psi = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix}$$

Dirac spinor

Covariant form of the Dirac equation

$$\hat{E} \psi = (\vec{\alpha} \hat{p} + \beta m) \psi$$

$$\Rightarrow i \frac{\partial \psi}{\partial t} = (-i \vec{\alpha} \vec{\nabla} + m \beta) \psi \quad (1)$$

introducing the 4 Dirac matrices

$$\gamma^0 = \beta \quad \gamma^1 = \beta \alpha_x \quad \gamma^2 = \beta \alpha_y \quad \gamma^3 = \beta \alpha_z$$

we multiply (1) by β (from the left)

$$i \beta \frac{\partial \psi}{\partial t} = (-i \beta \vec{\alpha} \vec{\nabla} + \beta^2 m) \psi$$

$$\Rightarrow i \gamma^0 \frac{\partial \psi}{\partial t} = (-i \vec{\gamma} \vec{\nabla} + m) \psi$$

We introduce $\gamma^\mu = (\gamma^0, \gamma^1, \gamma^2, \gamma^3)$

$$\partial_\mu = (\partial_t, \partial_x, \partial_y, \partial_z)$$

$$(i \gamma^\mu \partial_\mu - m) \psi = 0$$

reading
units

→
coming
back

from natural
units

$$(i \hbar \partial^\mu \partial_\mu - m c) \psi = 0$$

with Feynman slash notation

$$(i \not{\partial} - m) \psi = 0$$

$$\not{\partial} = \gamma^\mu \partial_\mu$$

$$\not{A} = \gamma^\mu A_\mu$$

Solutions of the equation

★ OK for spin-1/2 particles (fermions)

★ Solution can be $E > 0$ or $E < 0$

Feynman - Stueckelberg interpretation

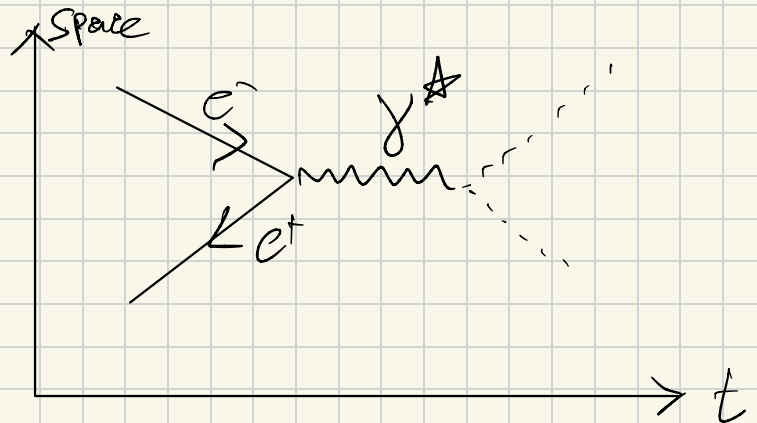
$E < 0$ solution = particle that goes back in time

$\Rightarrow (E > 0)$ antiparticles which flow in time

$E > 0$ solution = particle that moves fwd in time

Example: annihilation process

$$e^- e^+ \rightarrow \gamma^*$$



free spin-half spinor solutions

Particles $\psi_i = U_i e^{+i(\vec{P} \cdot \vec{x} - Et)}$

$$U_1(P) = \sqrt{E+m} \begin{pmatrix} 1 \\ 0 \\ \frac{P_z}{E+m} \\ \frac{P_x + iP_y}{E+m} \end{pmatrix}$$

$$U_2(P) = \sqrt{E+m} \begin{pmatrix} 0 \\ 1 \\ \frac{P_x - iP_y}{E+m} \\ -\frac{P_z}{E+m} \end{pmatrix}$$

Antiparticles $\psi_i = V_i e^{-i(\vec{P} \cdot \vec{x} - Et)}$ → antiparticle

$$V_1(P) = \sqrt{E+m} \begin{pmatrix} \frac{P_x - iP_y}{E+m} \\ -\frac{P_z}{E+m} \\ 0 \\ 1 \end{pmatrix}$$

$$V_2(P) = \sqrt{E+m} \begin{pmatrix} \frac{P_z}{E+m} \\ P_x + iP_y \\ 1 \\ 0 \end{pmatrix}$$

Compact - Form

$$u(s) = \begin{pmatrix} \phi^{(s)} \\ \frac{\vec{\sigma} \cdot \vec{p}}{E+m} \phi^{(s)} \end{pmatrix} \sqrt{E+m}$$

$$\phi^{(1)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \phi^{(2)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$
$$\chi^{(1)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \chi^{(2)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$v(s) = \begin{pmatrix} \frac{\vec{\sigma} \cdot \vec{p}}{E+m} \chi^{(s)} \\ \chi^{(s)} \end{pmatrix} \sqrt{E+m}$$

Completeness relation

$$\sum_{s=1}^2 u_{(s)} \bar{u}_{(s)} = (\gamma^\mu p_\mu + m) = (\not{p} + m)$$

$$\bar{u} = u^\dagger \gamma^0$$
$$= (\not{p}^\dagger)^T \gamma^0$$

Similarly for anti particles

$$\sum_{s=1}^2 v_{(s)} \bar{v}_{(s)} = \not{p} - m$$